

TAT-LLM: A Specialized Language Model for Discrete Reasoning over Financial Tabular and Textual Data

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Abstract

In this work, we develop a specialized language model with strong discrete reasoning capabilities to tackle question answering (QA) over hybrid tabular and textual data in finance. Compared with adopting online LLMs, specializing smaller LLMs is more advantageous in response to users' concerns about cost, network latency, and data security risks. To this end, we first abstract a *Step-wise Pipeline* for tabular and textual QA to help LLMs better execute multi-step inference, which includes three key steps, i.e. *Extractor*, *Reasoner* and *Executor*. This pipeline is proved to bring great performance gains compared with applying other prompting strategies like Chain-of-Thought (CoT), and meanwhile provides better interpretability to the derivation of the answer, with fixed inference steps and intermediate outcomes as references. We then develop a TAT-LLM model by fine-tuning LLaMA 2 with the training data generated automatically from existing datasets following the *Step-wise Pipeline*. The experimental results have verified that our TAT-LLM model can outperform all compared models, including prior best fine-tuned models and very large-scale LLMs like GPT-4 on FinQA, TAT-QA and TAT-DQA benchmarks. It is hoped that this work will shed light on practical solutions to the intelligent understanding of financial documents in the future. The generated datasets and trained models will be made publicly available to facilitate future research on the development of financial LLMs.

CCS Concepts

• Information systems → Question answering; • Computing methodologies → Natural language processing.

Keywords

Question Answering, Large Language Model, Tabular and Textual Data, Discrete Reasoning, Natural Language Processing

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1 Introduction

Documents containing both tables and text are ubiquitous in the financial industry, e.g. SEC filings and analyst reports, which often feature semantically intertwined tabular and textual contents as well as extensive numerical values. The intelligent understanding of these documents is generally studied with question answering (QA) tasks [3, 38], where discrete reasoning is indispensable. As exemplified in Figure 1, for a snippet from a real-world financial report containing both the table and text, the QA model needs to extract evidences from different parts to perform discrete reasoning to answer a given question, including but not limited to arithmetic calculations, comparing, counting, etc. A straightforward approach [17] is taking the table, text, and question as input and generating the answer end-to-end, which may be less ineffective due to the complex reasoning process involved [24]. Moreover, it is unclear how the answer has been inferred due to the lack of intermediate reasoning steps. Some methods [10, 36, 39] decompose the task into multiple steps and produce intermediate results as references, offering better interpretability. However, these methods often design distinct modules at each step, which need to be optimized concurrently through multi-task learning [33].

Recently, large language models (LLMs), e.g. GPT-4 [15] and FLAN [22], have exhibited strong reasoning abilities [23] with proper instructions, like Chain-of-Thought (CoT) [24] and least-to-most [35]. We are inspired to harness LLMs to facilitate discrete reasoning over hybrid tabular and textual data in the finance domain. We initially attempt to instruct LLMs with various prompting strategies such as CoT to complete the tasks, leading to unsatisfactory performance (Section 4.2) with a series of unpredictable steps. Then, we design a fixed *Step-wise Pipeline* as shown in Figure 2 b) and validate it on various LLMs including Vicuna [5], LLaMA 2-Chat [20] and GPT-3.5-Turbo [2] with FinQA [3], yielding promising results (Table 1). This *Step-wise Pipeline* is abstracted from classic

Revenue from external customers, classified by significant product and service offerings, was as follows:			
(in millions)			
Year Ended June 30,	2019	2018	2017
Server products and cloud services	32,622	26,129	21,649
Office products and cloud services	31,769	28,316	25,573
Windows	20,395	19,518	18,593
Gaming	11,386	10,353	9,051
Search advertising	7,628	7,012	6,219
LinkedIn	6,754	5,259	2,271
Enterprise Services	6,124	5,846	5,542
Devices	6,095	5,134	5,062
Other	3,070	2,793	2,611
Total	125,843	110,360	96,571
Our commercial cloud revenue, which includes Office 365, Commercial, Azure, the commercial portion of LinkedIn, Dynamics 365, and other commercial cloud properties, was \$38.1 billion, \$26.6 billion and \$16.2 billion in fiscal years 2019, 2018, and 2017, respectively. These amounts are primarily included in Office products and cloud services, Server products and cloud services, and LinkedIn in the table above.			
Question 1: What was the percentage change in gaming between 2018 and 2019?			
Discrete Reasoning Type: <u>Arithmetic Calculation</u>			
Answer: $(11,386 - 10,353) / 10,353 = 9.98\%$			
Question 2: Did the total revenue in 2019 exceed that of 2018?			
Discrete Reasoning Type: <u>Comparison</u>			
Answer: $125,843 > 110,360 = \text{Yes}$			
Question 3: How many revenue items are between 6,000 million and 6,500 million in 2019?			
Discrete Reasoning Type: <u>Counting</u>			
Answer: $\text{Count}(\text{Enterprise Services} \# \text{Devices}) = 2$			

Figure 1: Examples of QA with discrete reasoning over a hybrid of tabular and textual data.

multi-step methods for tabular and textual QA [36, 38, 39], consisting of 1) *Extractor* that identifies relevant information or evidence from given context; 2) *Reasoner* that generates a mathematical equation or logic rule with obtained information; and 3) *Executor* that executes the mathematical equation or logic rule with associated information to derive an answer. See Figure 2 b) for an illustration.

However, in real-world financial application scenarios, utilizing an online LLM tends to incur users' concerns on data security, not to mention the cost and network latency issues. By contrast, fine-tuning a smaller language model, such as the LLMs specifically for math word problems [6, 8], is fairly appealing and drawing increasing attention in the finance domain [27, 29, 31]. Hence, in pursuit of more practical solutions, we focus on specializing smaller language models to tackle the challenge following the *Step-wise Pipeline*. In particular, we develop a TAT-LLM model by fine-tuning LLaMA 2 [20] with the training data transformed automatically from existing expert-annotated financial tabular and textual QA datasets, as shown in Figure 2 c).

Specifically, we train our TAT-LLM model to take the table, text, and question as input, and complete the task in three key steps following the *Step-wise Pipeline*. We construct each of the training instances with five parts: Instruction, Table, Text, Question, and Response, from available tabular and textual QA datasets. Inspired by Tab-CoT [40], the model output (or response) is formatted to a structured table, with each row corresponding to one step of the

Prompting	Vicuna(7B)	LLaMA 2-Chat(7B)	GPT-3.5-Turbo
End-to-end	3.31	10.81	34.52
CoT	3.49	12.64	32.78
Step-wise (Ours)	10.11	15.43	54.23

Table 1: Accuracy of applying different zero-shot prompting methods to multiple LLMs on the test set of FinQA [3]. It can be seen that our abstracted *Step-wise Pipeline* beats end-to-end and CoT prompting methods.

pipeline, as shown in the right part of Figure 2 c). This design is friendly for further refining the outputs at each step and automatic evaluation. Moreover, to enhance the model performance, we equip the TAT-LLM model with an *External Executor*, which strengthens the execution of logical rules and mathematical calculations to better infer the final answer. We test our TAT-LLM model on three widely used benchmarks, including FinQA [3], TAT-QA [38] and TAT-DQA [37]. The experimental results show that our smallest model TAT-LLM (7B) can outperform all baseline models and even beat GPT-4 on all three datasets.

To summarize, we make the following main contributions:

- We abstract a fixed *Step-wise Pipeline* from classic previous works to instruct the LLMs, which not only brings better interpretability to the deviation of the answer, but also assists the LLMs in better performing discrete reasoning over hybrid tabular and textual data in finance.
- We develop a TAT-LLM model by fine-tuning LLaMA 2 (including 7B, 13B and 70B) for financial tabular and textual QA following the *Step-wise Pipeline*, better supporting practical application with cost and privacy concerns.
- We conduct extensive experiments on three popular benchmarks, validating the superiority of our TAT-LLM model over both conventional methods and very large-scale LLMs, e.g. GPT-4.

2 Related Work

2.1 Financial Tabular and Textual QA

Recent years have witnessed growing interest in developing QA models for the intelligent understanding of financial tabular and textual data, where discrete reasoning is indispensable as massive numerical content is usually included in the tables or text [3, 4, 7, 11, 34, 37, 38]. TAT-QA [38] and FinQA [3] are the pioneering studies and benchmarks on this challenge, which focus on QA over a table and a list of associated textual paragraphs. Later, TAT-HQA [11], PACIFIC [7] and TAT-DQA [37] datasets further extend TAT-QA to address counterfactual QA, proactive conversational QA and visually-rich document QA, respectively. In terms of methodology, lots of supervised methods are proposed to identify supporting evidences from the table and text first and infer the answer based on the extracted evidence [10, 12, 25, 25, 32, 36, 38, 39]. To name a few, TagOp [38] first adopts sequence tagging to extract supporting evidence from the table and text, and then applies pre-defined operators like addition, subtraction, and counting to infer the final answer; SoarGraph [39] extracts relevant information with table-text graphs and employs a tree-based method to generate the mathematical expression of the answer. Though LLMs quickly gain widespread popularity in numerous applications, their application to financial tabular and textual QA remains a relatively underexplored area, which is the main focus of our work.

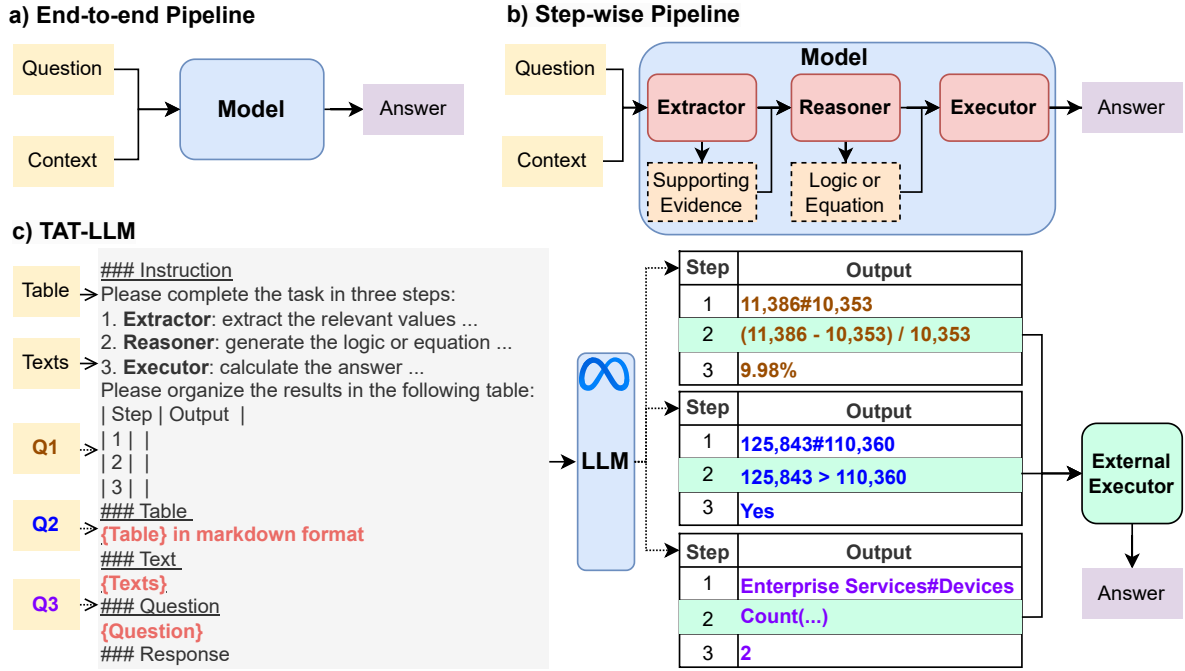


Figure 2: Illustration of a) *End-to-end Pipeline* and b) *Step-wise Pipeline*. c) Our TAT-LLM model is developed by fine-tuning LLaMA 2 following the *Step-wise Pipeline*.

2.2 Financial Large Language Models

The remarkable success of large language models (LLMs) like OpenAI ChatGPT [16] and GPT-4 [15] also benefits the financial industry [13]. In recent years, many financial LLMs have been proposed [1, 21, 26–29, 31]. BloombergGPT [26] is the first LLM in the finance domain, which is trained from scratch with Bloomberg’s proprietary data and used internally. Many financial LLMs like Instruct-FinGPT [31], FinMA [27], InvestLM [29] and FinVisGPT [21] are obtained by fine-tuning publicly available LLMs like LLaMA [19] with curated instructions. For instance, Instruct-FinGPT [31] is developed by fine-tuning LLaMA [19] with the instruction transformed from supervised financial sentiment analysis data, achieving impressive advancements in financial sentiment analysis; InvestLM [29] is trained on a small set of manually created instruction data to provide investment advice to financial practitioners with its strong capabilities in understanding financial text; FinVisGPT [21] is a multi-modal LLM specifically designed for financial chart analysis. In this work, we develop a small LLM (e.g., 7B) that specializes in reasoning over a hybrid of tabular and textual data in finance by fine-tuning LLaMA 2 [20] with instruction data automatically transformed from existing financial tabular and textual QA benchmarks.

3 Approach

In this section, we first explain the *Step-wise Pipeline* for tabular and textual QA in finance, and then elaborate on our TAT-LLM model.

3.1 Step-wise Pipeline

Large language models (LLMs) like ChatGPT [16] and FLAN [22] have demonstrated astoundingly strong multi-step reasoning abilities [23] following human instructions in many tasks [6, 35]. To better leverage such amazing power of LLMs to address QA over hybrid tabular and textual data, we abstract a *Step-wise Pipeline* from the previous multi-step methods [36, 38] on this task. As shown in Figure 2 b), this pipeline addresses the task with three key steps: *Extractor*, *Reasoner*, and *Executor*. In particular, *Extractor* serves as an information extraction module that identifies the relevant snippets or segments of information to the question from the context; 2) *Reasoner* works as a “logic thinker” to generate the right logic rule or equation that leads to the right answer to the question; 3) *Executor* is responsible for deriving the final answer by performing the logic rule or executing the equation. Compared with an *End-to-end Pipeline* that provides little insight into how the answer is derived, as shown in Figure 2 a), such a *Step-wise Pipeline* is able to produce a sequence of intermediate results that serve as the references to the final answer, leading to better interpretability. Furthermore, this design enables us to boost each component (or some of them) within the pipeline to consequently acquire improved final performance. For example, in this work, we choose to strengthen the third step by adding an extra *Executor*, which brings noticeable performance gains as verified experimentally in Section 4.3.4.

To instruct LLMs to perform discrete reasoning over tabular and textual data following the *Step-wise Pipeline*, we carefully design a natural language instruction to include the three key steps in the *Step-wise Pipeline*. We validate it on Vicuna (7B) [5], LLaMA

Algorithm 1 : External Executor

Input O_1 : the output of *Extractor*; O_2 : the output of *Reasoner*; O_3 : the output of *Executor*; Q_t : the predicted question type.

```

1:  $answer \leftarrow O_3$ 
2: if  $O_2$  is a valid arithmetic equation then
3:    $answer \leftarrow round(eval(O_2), 4)$ 
4: else if “#” in  $O_2$  then # multiple values are separated with “#”
5:    $arr \leftarrow O_2.split(“#”)$ 
6:    $answer \leftarrow len(arr)$ 
7: else if “>” in  $O_2$  or “<” in  $O_2$  then
8:    $answer \leftarrow eval(O_2)$ 
9: else if  $O_2 = \text{“N.A.”}$  then
10:  if  $Q_t = \text{“Span”}$  then
11:     $answer \leftarrow O_1$ 
12:  else if  $Q_t = \text{“Multiple Spans”}$  then
13:     $arr \leftarrow O_1.split(“#”)$  # spans are separated with “#”
14:     $answer \leftarrow arr$ 
15:  end if
16: end if

```

2-Chat (7B) [20] and GPT-3.5-Turbo [2] with FinQA [3] and obtain promising results (as shown in Table 1), demonstrating the rationality of applying the *Step-wise Pipeline* grounded on the LLMs. We then generate training data with publicly available tabular and textual QA datasets in finance and specialize a smaller LLM, i.e., LLaMA 2 [20], to answer questions following the *Step-wise Pipeline* for better supporting practical applications with cost and privacy concerns, as elaborated at below.

3.2 TAT-LLM

3.2.1 Selection of Base Language Model. We develop our TAT-LLM by fine-tuning LLaMA 2 [20]. LLaMA [19] is a series of open-source LLMs trained on the large-scale publicly available corpus. It is widely employed as the base LLM to develop other large language models with instruction tuning, such as Alpaca [18] and Vicuna [5]. In this study, we choose LLaMA 2 [20] as our base language model considering it offers an extended maximum sequence length (i.e., 4,096) compared to the previous version LLaMA [19], which often fails to accommodate the tabular and textual content of one instance from existing datasets. Moreover, LLaMA 2 has demonstrated remarkable performance across various benchmarks [20].

3.2.2 Construction of Training Data. Our fine-tuning is based on the training data we automatically transform from existing financial tabular and textual QA datasets. For implementation, we use three datasets, i.e., FinQA [3], TAT-QA [38] and TAT-DQA [37]. We carefully design templates for each dataset to generate the training data for fine-tuning LLMs. As shown in Figure 2 c), each training instance is designed to be composed of five parts, including Instruction, Table, Text, Question, and Response.

• **Instruction:** This part provides a detailed guide for what the task is and how to complete it. It outlines the steps that should be followed to derive the answer from the provided context. We adopt the *Step-wise Pipeline* described in above sections. 1) *Extractor*: We request the model to extract the relevant information from either the table or the accompanying text. 2) *Reasoner*: Based on the extracted

Dataset	Train Set	Dev Set	Test Set
FinQA	6,251	883	1,147
TAT-QA	13,215	1,668	1,669
TAT-DQA	13,251	1,645	1,662

Table 2: Statistics of the train, validation and test splits for the financial tabular and textual QA datasets.

information, we ask the model to derive an equation or a logic rule that is used to infer the answer. 3) *Executor*: We require the model to execute the equation or logic rule to arrive at the final answer. An illustration is given in Figure 2 c), which is the same for all the training instances. The instruction part helps the model to precisely understand the task and generate accurate and meaningful outputs.

• **Table:** This part refers to the tabular data in the input context of this task. We compose this part using the tables from FinQA and TAT-QA datasets. As each table in the two datasets is stored in a Two-Dimensional (2D) array, we transform it into a markdown table. Note that we omit TAT-DQA dataset for this part, considering the context given in TAT-DQA is the document pages from PDF files, where the structure of the table is unknown.

• **Text:** This part refers to the textual data in the input context. For FinQA and TAT-QA datasets, we compose this part with the paragraphs relevant to the table; all contents from the document pages in TAT-DQA dataset are regarded as textual data.

• **Question:** This part refers to the question that is asked based on the given tabular and textual context. We directly put the question from the three datasets in this part. The goal of the task is to generate the right answer to this question.

• **Response:** This part describes the model’s output based on the given table, text, and question. To better inspect the results in the intermediate steps and automatically evaluate the model output, we train the language model to format the output in a structured markdown table with two columns, i.e., step and output. Each row in the output table corresponds to one step defined in the instruction. In the end, the model draws its conclusion with a statement: “The answer is: {answer}”, making its final response clear.

To train the model, the response part in the training set needs to be generated in advance. Existing tabular and textual QA datasets often provide the answer to the question and the derivation annotated by human experts. To generate the correct response for constructing our training instance, we first automatically identify the supporting evidence from the derivation of each question, such as the numbers and entities, which are used as the output of the first step. We then convert the derivation to a valid equation or logic rule as the output of the second step, making sure its execution result is consistent with the final answer. We directly use the annotated answer as the final answer, which is also the output of the last step.

3.2.3 Training. After constructing the training data, we train our TAT-LLM model in various sizes, including 7B, 13B, and 70B, by fine-tuning LLaMA 2 [20] using Low-Rank Adaptation (LoRA) [9].

3.2.4 External Executor. We observe that the trained model struggles with the execution of the mathematical equations and logic rules (e.g., counting, comparison) in the last step, according to experimental evaluations in Section 4.3.4. Such incompetence of the

Type	Model	EM
Human Expert Performance		91.16
Fine-tuned	Longformer	21.90
	NeRd	48.57
	FinQANet _{BERT}	50.00
	DyRRen _{BERT}	59.37
	FinQANet _{RoBERTa}	61.24
	ELASTIC _{RoBERTa}	62.66
	DyRRen _{RoBERTa}	63.30
Zero-shot	FinMA (7B)	4.00
	Vicuna (7B)	10.11
	FinMA (30B)	11.00
	LLaMA 2-Chat (7B)	15.43
	InvestLM (65B)	29.00
	LLaMA 2-Chat (70B)	32.17
	MAmmoTH (70B)	36.09
	WizardMath (70B)	47.25
	GPT3.5-Turbo	58.00
	GPT-4	<u>63.91</u>
Ours	TAT-LLM (7B)	64.60

Table 3: Performance of our TAT-LLM model and compared models on the test set of FinQA. The best are marked in bold and the second-best are underlined.

Executor has also been found in previous works [14, 30]. To enhance the model’s accuracy of the final output, we propose to add an *External Executor* to our model, which takes the model’s intermediate outputs as input and performs the execution of the equations or logic rules to obtain the final answer. With this *External Executor*, the model can refine its output to a better one. With the output of the model formatted to a structured table, we can easily access the intermediate results. After obtaining the intermediate results, the *External Executor* is applied to refine the final answer instead of directly using the prediction of the model. The whole process is summarized in Algorithm 1.

4 Experiments

4.1 Datasets, Models and Evaluation Metrics

4.1.1 Datasets. We use below three datasets for our experiments. See Table 2 for statistics of the splits on each dataset.

- **FinQA** [3] is an expert-annotated tabular and textual QA dataset in which the tables and text are sampled from financial reports. It focuses on discrete reasoning capabilities like addition, subtraction, multiplication, division, and numerical comparison.
- **TAT-QA** [38] is also built with tables and paragraphs extracted from financial reports. Most questions require discrete reasoning for inferring the answers; meanwhile, there are also cases where answers can be extracted directly from the tables or text. Its questions are classified into four different types: *Span*, *Multiple Spans*, *Counting*, and *Arithmetic*.
- **TAT-DQA** [37] is an extension of the TAT-QA dataset, focusing on question answering over original long financial statements with up to three pages, where table positions and structures are unknown.

Type	Model	EM	F ₁
Human Expert Performance		84.1	90.8
Fine-tuned	TagOp	50.10	58.00
	TeaBReaC	55.80	63.80
	KIQA	58.20	67.40
	FinMath	58.30	68.20
	GANO	61.90	72.10
	MHST	63.60	72.70
	UniPCQA	63.90	72.20
	SoarGraph	65.40	75.30
	UniRPG	67.20	76.00
	RegHNT	70.30	77.90
	MVGE	70.90	79.10
Zero-shot	Vicuna (7B)	32.53	40.97
	LLaMA 2-Chat (7B)	37.16	45.37
	MAmmoTH (70B)	38.97	46.51
	WizardMath (70B)	39.63	45.28
	LLaMA 2-Chat (70B)	45.94	53.80
	GPT3.5-Turbo	59.47	68.11
	GPT-4	<u>71.92</u>	<u>79.71</u>
Ours	TAT-LLM (7B)	74.56	82.88

Table 4: Performance of our TAT-LLM model and compared models on the test set of TAT-QA. Best results are marked in bold and second-best results are underlined.

4.1.2 Compared Models. We compare our TAT-LLM model with two groups of models: fine-tuned models on the tabular and textual QA dataset, and LLMs in both zero-shot and few-shot settings. For fine-tuned models, we select the state-of-the-art supervised fine-tuned models for each dataset, which are listed in Table 3, Table 4 and Table 5 respectively. For LLMs, we compare our models with the general LLMs, including the state-of-the-art GPT3.5-Turbo [2] and GPT-4 [15], and the powerful smaller LLMs Vicuna (7B) [5] and LLaMA 2-Chat (7B and 70B) [20], which have garnered considerable research attention recently. In addition, we also include the LLMs specialized in math word problems into comparisons with our models, including MAmmoTH (70B) [30] and WizardMath (70B) [14], as our task involves numerical reasoning, as well as the financial large language models, including FinMA [27] and InvestLM [29].

4.1.3 Evaluation Metrics. For TAT-QA and TAT-DQA datasets, we adopt the Exact Match (EM) and the numeracy-focused (macro-averaged) F₁ score [37, 38]. Both metrics measure the overlap between a bag-of-words representation of the gold and predicted answers. The numeracy-focused F₁ score is set to 0 unless the predicted number is exactly equal to the ground truth. We use EM for FinQA, which is the same as the metric of Execution Accuracy originally used in FinQA [3].

For zero-shot and few-shot prediction using LLMs, we omit the scale prediction and compare the predicted value with the ground truth value only, assuming the scale prediction is always correct. However, for our TAT-LLM model, since we instruct the model to output a structured table that is friendly to automatic evaluation, we take the scale into account.

Type	Model	EM	F ₁
Human Expert Performance		84.1	90.8
Fine-tuned	NumNet+ V2	30.60	40.10
	TagOp	33.70	42.50
	MHST	41.50	50.70
	Doc2SoarGraph	59.20	67.60
Zero-shot	Vicuna (7B)	28.44	36.72
	LLaMA 2-Chat (7B)	34.52	42.32
	MAmmoTH (70B)	35.42	42.82
	WizardMath (70B)	36.44	41.55
	LLaMA 2-Chat (70B)	41.91	49.74
	GPT3.5-Turbo	52.74	61.40
	GPT-4	<u>64.46</u>	<u>72.20</u>
Ours	TAT-LLM (7B)	69.45	77.75

Table 5: Performance of our TAT-LLM model and compared models on the test set of TAT-DQA. Best results are marked in bold and second-best results are underlined.

4.2 Main Results

We first compare the performance of our TAT-LLM with previous methods on each dataset respectively. The experimental results are summarized in Table 3 for FinQA, Table 4 for TAT-QA and Table 5 for TAT-DQA. From the tables, we make the following observations. 1) Our TAT-LLM (7B) significantly outperforms all the previous models on each of the three datasets, including the previous best fine-tuned models and the state-of-the-art GPT-4 [15]. In particular, our TAT-LLM reaches 64.60%, 74.56% and 69.45% in terms of EM on the test set of FinQA, TAT-QA and TAT-DQA respectively, i.e. an increase of 0.69, 2.64 and 4.99 points compared to GPT-4. These results well demonstrate the noticeable effectiveness of our TAT-LLM model. The results also confirm the rousing potential of specializing smaller language models, like our TAT-LLM (7B) model, for specific tasks to yield better performance than very large-scale models like GPT-4. 2) However, the performance of GPT-4 and our TAT-LLM (7B) obviously lags behind that of human experts, showing that this task is still challenging. 3) The strong general LLM GPT3.5-Turbo [16] and LLaMA 2-Chat (70B) [20] underperform the best fine-tuned models, evidencing that supervised fine-tuning is still essential for achieving advanced performance for this challenge. 4) The LLMs specialized in mathematical reasoning, i.e., WizardMath [14] and MAmmoTH [30], largely underperform GPT3.5-Turbo, GPT-4 and our TAT-LLM model, showing that current numerically-enhanced LLMs still struggle with discrete

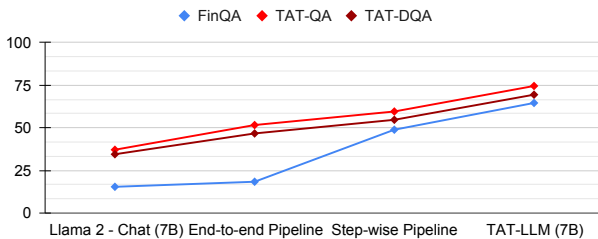


Figure 3: Comparison of different training strategies in terms of Exact Match on all three datasets.

Model	FinQA	TAT-QA		TAT-DQA	
	EM	EM	F ₁	EM	F ₁
Zero-shot prompting:					
LLaMA 2-Chat (70B)	32.17	45.94	53.80	41.91	49.74
GPT-3.5-Turbo	58.00	59.47	68.11	52.74	61.40
GPT-4	63.91	71.92	79.71	64.46	72.20
Few-shot prompting (3-shot):					
LLaMA 2-Chat (70B)	34.52	35.72	45.50	35.72	45.50
GPT-3.5-Turbo	59.37	51.47	62.21	55.74	64.49
GPT-4	76.55	63.38	75.27	68.67	76.92
Fine-tuned with respective training set :					
TAT-LLM (7B)	64.60	74.56	82.88	69.45	77.75
Fine-tuned with a combination of training sets:					
TAT-LLM _{All} (7B)	65.13	76.49	85.13	71.38	80.24
TAT-LLM _{All} (13B)	71.93	77.51	85.95	72.22	80.56
TAT-LLM _{All} (70B)	76.81	81.42	88.49	76.55	83.90

Table 6: Performance of our TAT-LLM model trained with the combination of all three train sets, and representative baseline LLMs using zero-shot and few-shot prompting. Best results are marked in bold.

reasoning over tabular and textual QA. This again speaks for the considerable value of specializing models for specific tasks.

4.3 In-Depth Analysis

4.3.1 Fine-tuning with Combined Data. In addition to training our TAT-LLM model on a single tabular and textual QA dataset, we also train the TAT-LLM_{All} (7B) with a combination of train sets from FinQA, TAT-QA and TAT-DQA datasets. As shown in Table 6, training on a combined dataset brings obvious performance improvements over training on a single dataset. This suggests that incorporating larger-sized and more diverse data into the training process can potentially enhance the model’s overall capabilities. Furthermore, we check the performance with various sizes of the base LLaMA 2 model, and find that larger models consistently achieve significant performance improvements across all three datasets, aligning with the prevailing consensus in the field.

Model	FinQA	TAT-QA		TAT-DQA	
	EM	EM	F ₁	EM	F ₁
GPT-4	63.91	71.92	79.71	64.46	72.20
TAT-LLM _{All} (7B)					
w/o External Executor	48.47	58.69	67.21	54.84	63.68
w External Executor	65.13	76.49	85.13	71.38	80.24
gains (+)	16.66	17.80	17.92	16.54	16.56
TAT-LLM _{All} (13B)					
w/o External Executor	60.05	62.60	70.73	59.95	68.61
w External Executor	71.75	76.79	85.05	71.86	80.50
gains (+)	11.70	14.19	14.32	11.91	11.89
TAT-LLM _{All} (70B)					
w/o External Executor	70.10	76.61	83.55	71.74	78.99
w External Executor	76.81	81.42	88.49	76.55	83.90
gains (+)	6.71	4.81	4.94	4.81	4.91

Table 7: Effectiveness of the External Executor for TAT-LLM_{All} with different sizes.

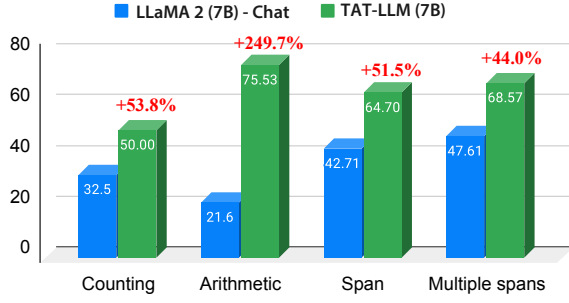


Figure 4: Performance comparison in terms of EM between TAT-LLM (7B) and LLaMA 2-Chat (7B) for different question types on TAT-DQA.

4.3.2 Comparison with Different Prompting Methods. We compare our TAT-LLM model with three advanced LLMs, i.e. LLaMA 2-Chat (70B), GPT-3.5-Turbo, and GPT-4, in both zero-shot and few-shot settings (3-shot). For each setting, we apply three different prompting methods, including *End-to-end*, Chain-of-Thought (CoT), and our *Step-wise Pipeline*, to instruct LLMs to answer the questions. We report the best results in Table 6, from which we make three observations. 1) Our fine-tuned TAT-LLM_{ALL} (70B) model outperforms all other LLMs, including the advanced GPT-4, in both zero-shot and few-shot settings, indicating the necessity of specializing models for addressing this challenge. 2) Applying few-shot prompting on baseline LLMs brings significant performance increase on FinQA, but the performance is still worse than our TAT-LLM (70B) model. 3) However, performance drop can be observed on TAT-QA and TAT-DQA for some LLMs after applying few-shot prompting compared to that with zero-shot prompting applied. This is likely because both datasets contain four types of questions, and randomly selecting examples for few-shot prompting may confuse the LLMs.

4.3.3 Analysis on Fine-tuning Strategies. To verify the effect of our fine-tuning strategy, i.e. *Step-wise Pipeline* plus *External Executor*, we compare it with separately applying *End-to-end Pipeline* or *Step-wise Pipeline* on LLaMA 2 (7B) models. The results are summarized in Figure 3. 1) We can see our TAT-LLM (7B) clearly outperforms the two variants following the *End-to-end Pipeline* and the *Step-wise Pipeline* as well as the LLaMA 2-Chat (7B) model on all three financial tabular and textual QA datasets, demonstrating the effectiveness of our fine-tuning strategy. 2) We find that the *Step-wise Pipeline* also outperforms the *End-to-end Pipeline* in all situations, showing the rationality of incorporating intermediate reasoning steps, which is in line with previous research on chain-of-thoughts reasoning [24]. 3) It is observed that the *End-to-end Pipeline* achieves notably poor performance on FinQA compared to that on TAT-QA and TAT-DQA. This is probably because FinQA requires more complex discrete reasoning steps than TAT-QA and TAT-DQA, and omitting such steps in fining-tuning would largely degrade model performance. 4) We find a noticeable performance increase from the *Step-wise Pipeline* to TAT-LLM. This is due to the incorporation of the reliable *External Executor* which largely guarantees the reasoning accuracy of the answer, which is analyzed at below.

4.3.4 Effect of External Executor w.r.t. Different Model Sizes. Here we analyze the effectiveness of the *External Executor* on models

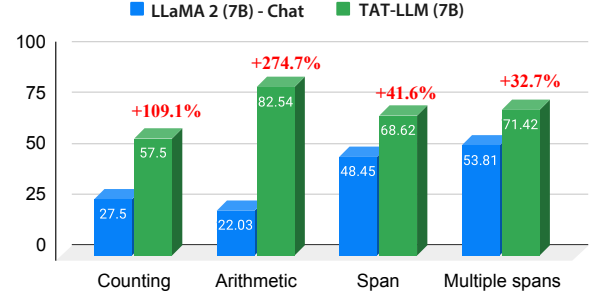


Figure 5: Performance comparison in terms of EM between TAT-LLM (7B) and LLaMA 2-Chat (7B) for different question types on TAT-QA.

with different sizes. According to the experimental results shown in Table 7, we can see that ablating the *External Executor* significantly degrades the performance across all model sizes on all datasets, causing decreases of larger than 15% on the EM of the three datasets for TAT-LLM_{ALL} (7B). This shows that the *External Executor* is essential in ensuring the answer correctness. Also, we find that larger models w/o *External Executor* perform better than smaller models w/o *External Executor*, and the gains by the *External Executor* get smaller as the model size increases. The largest TAT-LLM_{ALL} (70B) w/o *External Executor* only decreases less than 7% on all three datasets, which is far less than TAT-LLM_{ALL} (7B). This is because larger models have stronger discrete reasoning abilities than smaller models and make less mistakes in answer calculation. Hence, the advantage of ensuring accurate results in discrete reasoning through the *External Executor* appears to be less conspicuous.

4.3.5 Improvements w.r.t. Different Question Types. Here we analyze the improvements achieved by TAT-LLM model regarding different question types of TAT-QA and TAT-DQA datasets. We compare the performance of TAT-LLM (7B) and LLaMA 2-Chat (7B) for each question type. The results are summarized in Figure 5 and Figure 4 for TAT-QA and TAT-DQA, respectively. We can observe substantial performance improvements for the *Arithmetic* questions, which are over two-fold, for both datasets. The larger performance gains are attributed to the effective fine-tuning strategy of TAT-LLM as well as the reliable *External Executor* that ensures accurate execution of the equations and logic rules. Large performance improvements are also found for the *Counting* questions, namely 109.1% for TAT-QA and 53.8% for TAT-DQA. This is probably due to executing more correct counting operations. For the *Span* and *Multiple Spans* types, relatively smaller improvements are observed, which are still over 30%. This is potentially because these questions involve simpler or no discrete reasoning, mainly numerical comparison.

5 Conclusion

In this work, we abstract a *Step-wise Pipeline* for tabular and textual QA based on the previous classic multi-step methods, and develop a TAT-LLM model by specializing smaller language models (i.e., 7B) to perform discrete reasoning over financial tabular and textual data. We validate its effectiveness with extensive experiments, showing that TAT-LLM can outperform both conventional methods and very

large-scale LLMs like ChatGPT and GPT-4 on this task. Our work well demonstrates that specialized models might be a promising direction towards more advanced models on specific tasks that can compete with human experts.

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